

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		Substitution: $W = F \times d = 50\,000 \times 20$ (1) $W = 1\,000\,000$ [J] (1)	1	1		2	2	
		(ii)		[Energy transfer = $1\,000\,000$ (ecf) – $600\,000$] = $400\,000$ [J] N.B. Don't award the ecf mark if negative energy calculated or if subtracted incorrectly		1		1	1	
	(b)			Any 2 × 1: Make it [more] streamlined / using materials in tyres which don't heat up [as much] or lower profile tyres / stop – start systems / use lighter materials / automatic tyre pressure warnings / eco system / regenerative braking or hybrid or electric cars Don't accept smaller car or smaller engine	2			2		
				Question 2 total	3	2	0	5	3	0